

INTRODUCTION

That light should have mechanical properties has been known, or at least suspected, since Kepler proposed that the tails of comets were due to radiation pressure associated with light from the sun. A quantitative theory of such effects became possible only after the development of Maxwell's unified theory of electricity, magnetism and optics. However, although his treatise on electromagnetism (1.1) contains a calculation of the radiation pressure at the earth's surface, there is little more on the mechanical effects of light. It was Poynting who quantified the momentum and energy flux associated with an electromagnetic field (1.2). In modern terms, the momentum per unit volume associated with an electromagnetic wave is given by $\epsilon_0 \mathbf{E} \times \mathbf{B}$. The angular momentum density is, naturally enough, the cross product of this with position, that is $\mathbf{r} \times \epsilon_0 (\mathbf{E} \times \mathbf{B})$ (1.3).

Poynting reasoned that circularly polarised light must carry angular momentum (1.4, **Paper 1.1**). His argument proceeded by analogy with the wave motion associated with a line of dots marked on a rotating cylindrical shaft. His calculation showed that $E\lambda/2\pi$ is the angular momentum transmitted through a plane in unit time, per unit area, where E is Poynting's notation for the energy per unit volume and λ is the wavelength. When the energy of each photon crossing the surface is associated with $\hbar\omega$, we obtain the result that circularly polarised photons each carry $\hbar$ units of angular momentum.

Poynting's paper concludes with a proposal for measuring the angular momentum associated with circularly polarised light. His idea was that circularly polarised light passing through a large number of suspended quarter-wave plates, and so becoming linearly polarised, should give up all its angular momentum and so induce a rotation in the suspension. He concludes, however, that "my present experience of light-forces does not give me much hope that the effect could be detected". The effect was detected, however, about twenty years after Poynting's death by Beth (1.5, **Paper 1.2**) who used a single quarter wave plate, together with a mirror which sent the light back through the plate enhancing the torque on the suspension. He showed that the same quantitative result is obtained for the classical torque as for that which arises from the assumption that each photon carried an angular momentum $\hbar$.

Careful examination of $\epsilon_0 \mathbf{r} \times (\mathbf{E} \times \mathbf{B})$ shows that polarisation does not account for all of the angular momentum that can be carried by the electromagnetic field. The part associated with polarisation is known as spin, but in addition there is also an orbital contribution. Until recently, however, the discussion of spin and orbital angular momenta for light was largely restricted to textbooks and related to non-specific forms for the electric field. Current research activity in this area originated with the realisation that physically

realisable light beams, familiar from the paraxial optics of laser theory, can carry a well-defined quantity of orbital angular momentum for each photon (1.6, **Paper 2.1**). An extensive review of orbital angular momentum was published in 1999 (1.7). This was followed, in 2002, by the publication of a special issue of *Journal of Optics B* devoted to atoms and angular momentum of light. The introduction to that special issue provides a brief overview of the current status of the field and is reprinted here as the natural introduction to this book (1.8, **Paper 1.3**).

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The Wave Motion of a Revolving Shaft, and a Suggestion as to the Angular Momentum in a Beam of Circularly Polarised Light

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When a shaft of circular section is revolving uniformly, and is transmitting power uniformly, a row of particles originally in a line parallel to the axis will lie in a spiral of constant pitch, and the position of the shaft at any instant may be described by the position of this spiral.

Let us suppose that the power is transmitted from left to right, and that as viewed from the left the revolution is clockwise. Then the spiral is a left-handed screw. Let it be on the surface, and there make an angle ε with the axis. Let the radius of the shaft be a , and let one turn of the spiral have length λ along the axis. We may term λ the wave-length of the spiral. We have $\tan \varepsilon = 2\pi a/\lambda$. If the orientation of the section at the origin at time t is given by $\theta = 2\pi Nt$, where N is the number of revolutions per second, the orientation of the section at x is given by

$$\theta = 2\pi Nt - \frac{x}{a} \tan \varepsilon = \frac{2\pi}{\lambda} (N\lambda t - x), \quad (1)$$

which means movement of orientation from left to right with velocity $N\lambda$.

The equation of motion for twist waves on a shaft of circular section is

$$\frac{d^2\theta}{dt^2} = U_n^2 \frac{d^2\theta}{dx^2}, \quad (2)$$

where U_n^2 = modulus of rigidity/density = n/ρ .

Though (1) satisfies (2), it can hardly be termed a solution for $d^2\theta/dt^2$, and $d^2\theta/dx^2$ in (1) are both zero. But we may adapt a solution of (2) to fit (1) if we assume certain conditions in (1).

The periodic value

$$\theta = \Theta \sin \frac{2\pi}{l} (U_n t - x)$$

satisfies (2), and is a wave motion with velocity U_n and wave-length l . Make l so great that for any time or for any distance under observation $U_n t/l$ and x/l are so small that the angle may be put for the sine. Then

$$\theta = \Theta \frac{2\pi}{l} (U_n t - x). \quad (3)$$

This is uniform rotation. It means that we only deal with the part of the wave near a node, and that we make the wave-length l so great that for a long distance the “displacement curve” obtained by plotting θ against t coincides with the tangent at the node. We must distinguish, of course, between the wave-length l of the periodic motion and the wave-length λ of the spiral.

We can only make (1) coincide with (3) by putting

$$\Theta/l = 1/\lambda \quad \text{and} \quad N\lambda = U_n.$$

Then it follows that for a given value of N , the impressed speed of uniform rotation, there is only one value of λ or one value of ε for which the motion may be regarded as part of a natural wave system, transmitted by the elastic forces of the material with velocity $= \sqrt{(n/\rho)}$. There is therefore only one “natural” rate of transmission of energy.

The value of ε is given by

$$\tan \varepsilon = 2\pi a/\lambda = 2\pi aN/N\lambda = 2\pi aN/U_n = 2\pi aN\sqrt{(\rho/n)}.$$

Suppose, for instance, that a steel shaft with radius $a = 2$ cm, density $\rho = 7.8$, and rigidity $n = 10^{12}$ is making $N = 10$ revs. per sec. We may put $\tan \varepsilon = \varepsilon$, since it is very small. The shaft is twisted through 2π in length λ or through $2\pi/\lambda$ per centimetre, and the torque across a section is

$$G = \frac{1}{2} n\pi a^4 2\pi/\lambda = n\pi^2 a^4 N\sqrt{(\rho/n)},$$

since

$$\lambda = \frac{U_n}{N} = \frac{1}{N} \sqrt{\frac{n}{\rho}}.$$

The energy transmitted per second is

$$2\pi NG = 2\pi^3 a^4 N^2 \sqrt{(n\rho)}.$$

Putting 1 H.P. $= 746 \times 10^7$ ergs per second, this gives about 38 H.P.

But a shaft revolving with given speed N can transmit any power, subject to the limitation that the strain is not too great for the material. When the power is not that “naturally” transmitted, we must regard the waves as “forced.” The velocity of transmission is no longer U_n , and forces will have to be applied from outside in addition to the internal elastic forces to give the new velocity.

Let H be the couple applied per unit length from outside. Then the equation of motion becomes

$$\frac{d^2\theta}{dt^2} = U_n^2 \frac{d^2\theta}{dx^2} + \frac{2H}{\pi a^4},$$

where $\frac{1}{2}\pi a^4$ is the moment of inertia of the cross section. Assuming that the condition travels on with velocity U unchanged in form,

$$\frac{d\theta}{dt} = -U \frac{d\theta}{dx} \quad \text{and} \quad H = \frac{1}{2}\pi a^4 (U_n^2) \frac{d^2\theta}{dx^2},$$

or H has only to be applied where $d^2\theta/dx^2$ has value, that is where the twist is changing.

The following adaptation of Rankine's tube method of obtaining wave velocities* gives these results in a more direct manner. Suppose that the shaft is indefinitely extended both ways. Any twist disturbance may be propagated unchanged in form with any velocity we choose to assign, if we apply from outside the distribution of torque which, added to the torque due to strain, will make the change in twist required by the given wave motion travelling at the assigned speed.

Let the velocity of propagation be U from left to right, and let the displacement at any section be θ , positive if clockwise when seen from the left. The twist per unit length is

$$\frac{d\theta}{dx} = -\frac{1}{U} \frac{d\theta}{dt} = -\frac{\dot{\theta}}{U}.$$

The torque across a section from left to right in clockwise direction is

$$-\frac{1}{2}n\pi a^4 \frac{d\theta}{dx} = \frac{n\pi a^4}{2U} \cdot \dot{\theta}.$$

Let the shaft be moved from right to left with velocity U ; then the disturbance is fixed in space, and if we imagine two fixed planes drawn perpendicular to the axis, one, A , at a point where the disturbance is θ and the other, B , outside the wave system, where there is no disturbance, the condition between A and B remains constant, except that the matter undergoing that condition is changing. Hence the total angular momentum between A and B is constant. But no angular momentum enters at B , since the shaft is there untwisted and has merely linear motion. At A , then, there must be on the whole no transfer of angular momentum from right to left. Now, angular momentum is transferred in three ways:—

1. By the carriage by rotating matter. The angular momentum per unit length is $\frac{1}{2}\rho\pi a^4\dot{\theta}$, and since length U per second passes out at A , it carries out $\frac{1}{2}\rho\pi a^4\dot{\theta}U$.

2. By the torque exerted by matter on the right of A on matter on the left of A . This takes out $-n\pi a^4\dot{\theta}/2U$.

3. By the stream of angular momentum by which we may represent the forces applied from outside to make the velocity U instead of U_n .

If H is the couple applied per unit length, we may regard it as due to the flow of angular momentum L along the shaft from left to right, such that $H = -dL/dx$. There is then angular momentum L flowing out per second from right to left. Since the total flow due to (1), (2), and (3) is zero,

$$\frac{1}{2}\rho\pi a^4\dot{\theta}U - n\pi a^4\dot{\theta}/2U - L = 0,$$

and

$$L = \frac{\pi a^4 \dot{\theta}}{2} \left(\rho U - \frac{n}{U} \right) = \frac{\rho\pi a^4 \dot{\theta}}{2U} (U_n^2) = -\frac{\rho\pi a^4}{2} \frac{d\theta}{dx} (U^2 - U_n^2),$$

* 'Phil. Trans.,' 1870, p. 277.

and

$$H = -\frac{dL}{dx} = \frac{\rho\pi a^4}{2} \frac{d^2\theta}{dx^2} (U_n^2).$$

If $H = 0$, either $U^2 = U_n^2$ when the velocity has its “natural value,” or $d^2\theta/dx^2 = 0$, and the shaft is revolving with uniform twist in the part considered.

Now put on to the system a velocity U from left to right. The motion of the shaft parallel to its axis is reduced to zero, and the disturbance and the system H will travel on from left to right with velocity U . A “forced” velocity does not imply *transfer* of physical conditions by the material with that velocity. We can only regard the conditions as reproduced at successive points by the aid of external forces. We may illustrate this point by considering the incidence of a wave against a surface. If the angle of incidence is i and the velocity of the wave is V , the line of contact moves over the surface with velocity $v = V/\sin i$, which may have any value from V to infinity. The velocity v is not that of transmission by the material of the surface, but merely the velocity of a condition impressed on the surface from outside.

Probably in all cases of transmission with forced velocity, and certainly in the case here considered, the velocity depends upon the wave-length, and there is dispersion.

With a shaft revolving N times per second $U = N\lambda$, and it is interesting to note that the group velocity $U - \lambda dU/d\lambda$ is zero. It is not at once evident what the group velocity signifies in the case of uniform rotation. In ordinary cases it is the velocity of travel of the “beat” pattern, formed by two trains of slightly different frequencies. The complete “beat” pattern is contained between two successive points of agreement of phase of the two trains. In our case of superposition of two strain spirals with constant speed of rotation, points of agreement of phase are points of intersection of the two spirals. At such points the phases are the same, or one has gained on the other by 2π . Evidently as the shaft revolves these points remain in the same cross-section, and the group velocity is zero.

With deep water waves the group velocity is half the wave velocity, and the energy flow is half that required for the onward march of the waves.* The energy flow thus suffices for the onward march of the group, and the case suggests a simple relation between energy flow and group velocity.

But the simplicity is special to unforced trains of waves. Obviously, it does not hold when there are auxiliary working forces adding or subtracting energy along the waves. For the revolving shaft the simple relation would give us no energy flow, whereas the strain existing in the shaft implies transmission of energy at a rate given as follows.

The twist per unit length is $d\theta/dx$, and therefore the torque across a section is $-\frac{1}{2}n\pi a^4 d\theta/dx$, or $\frac{1}{2}n\pi a^4 \dot{\theta}/U$, since $d\theta/dx = -\dot{\theta}/U$. The rate of working or of energy flow across the section is $\frac{1}{2}n\pi a^4 \dot{\theta}^2/U$.

The relation of this to the strain and kinetic energy in the shaft is easily found. The strain energy per unit length being $\frac{1}{2}$ (couple $\times$ twist per unit length) is $\frac{1}{2}n\pi a^4 (d\theta/dx)^2$, which is $\frac{1}{4}n\pi a^4 \dot{\theta}^2/U^2$. The kinetic energy per unit length is $\frac{1}{2}\rho\pi a^4 \dot{\theta}^2$, or, putting $\rho = n/U_n^2$, is $\frac{1}{4}n\pi a^4 \dot{\theta}^2/U_n^2$.

In the case of natural velocity, for which no working forces along the shaft are needed, when $U = U_n = \sqrt{(n/\rho)}$, the kinetic energy is equal to the strain energy at every point and the energy transmitted across a section per second is that contained in length U_n .

*O. Reynolds, ‘Nature,’ August 23, 1877; Lord Rayleigh, ‘Theory of Sound,’ vol. 1, p. 477.

But if the velocity is forced this is no longer true,* and it is easily shown that the energy transferred is that in length

$$\frac{2U}{1 + U^2/U_n^2},$$

which is less than U if $U > U_n$, and is greater than U if $U < U_n$.

It appears possible that always the energy is transmitted along the shaft at the speed U_n . If the forced velocity $U > U_n$, we may, perhaps, regard the system in a special sense as a natural system with a uniform rotation superposed on it.

Let us suppose that the whole of the strain energy in length U_n is transferred per second while only the fraction μ of the kinetic energy is transferred, the fraction $1 - \mu$ being stationary.

The energy transferred: strain energy in U_n : kinetic energy in $U_n = 1/U : U_n/2U^2 : U_n/2U_n^2$.

Put $U = pU_n$, and our supposition gives

$$\frac{1}{pU_n} = \frac{1}{2p^2U_n} + \frac{\mu}{2U_n} \quad \text{or} \quad \mu = \frac{2}{p} - \frac{1}{p^2} = 1 - \left(1 - \frac{1}{p}\right)^2.$$

If the forced velocity $U < U_n$, we may regard the system as a natural one, with a uniform stationary strain superposed on it.

We now suppose that the whole of the kinetic energy is transferred, but only a fraction ν of the strain energy, and we obtain

$$\frac{1}{pU_n} = \frac{\nu}{2p^2U_n} + \frac{1}{2U_n} \quad \text{or} \quad \nu = 2p - p^2 = 1 - (1 - p)^2.$$

It is perhaps worthy of note that a uniform longitudinal flow of fluid may be conceived as a case of wave motion in a manner similar to that of the uniform rotation of a shaft.

A SUGGESTION AS TO THE ANGULAR MOMENTUM IN A BEAM OF CIRCULARLY POLARISED LIGHT

A uniformly revolving shaft serves as a mechanical model of a beam of circularly polarised light. The expression for the orientation θ of any section of the shaft distant x from the origin, $\theta = 2\pi\lambda^{-1}(Ut - x)$, serves also as an expression for the orientation of the disturbance, whatever its nature, constituting circularly polarised light.

For simplicity, take a shaft consisting of a thin cylindrical tube. Let the radius be a , the cross-section of the material s , the rigidity n , and the density ρ . Let the tube make N revolutions per second, and let it have such twist on it that the velocity of transmission of the spiral indicating the twist is the natural velocity $U_n = \sqrt{(n/\rho)}$.

Repeating for this special case what we have found above, the strain energy per unit length is $\frac{1}{2}n\varepsilon^2s$, or, since $\varepsilon = a d\theta/dx = -a\dot{\theta}/U_n$, the strain energy is $\frac{1}{2}na^2s\dot{\theta}^2/U_n^2 = \frac{1}{2}\rho a^2s\dot{\theta}^2$.

*In the Sellmeier model illustrating the dispersion of light, the particles may be regarded as outside the material transmitting the waves and as applying forces to the material which make the velocity forced. The simple relation between energy flow and group velocity probably does not hold for this model.

But the kinetic energy per unit length is also $\frac{1}{2}\rho a^2 s \dot{\theta}^2$, so that the total energy in length U_n is $\rho a^2 s \dot{\theta}^2 U_n$. The rate of working across a section is

$$n \varepsilon s a \dot{\theta} = n a^2 s \dot{\theta}^2 / U_n = \rho a^2 s \dot{\theta}^2 U_n,$$

or the energy transferred across a section is the energy contained in length U_n .

If we put E for the energy in unit volume and G for the torque per unit area, we have $G s \dot{\theta} = E s U_n$, whence

$$G = E U_n / \dot{\theta} = E N \lambda / 2\pi N = E \lambda / 2\pi.$$

The analogy between circularly polarised light and the mechanical model suggests that a similar relation between torque and energy may hold in a beam of such light incident normally on an absorbing surface. If so, a beam of wave-length λ containing energy E per unit volume will give up angular momentum $E \lambda / 2\pi$ per second per unit area. But in the case of light waves $E = P$, where P is the pressure exerted. We may therefore put the angular momentum delivered to unit area per second as

$$P \lambda / 2\pi.$$

In the 'Philosophical Magazine,' 1905, vol. 9, p. 397, I attempted to show that the analogy between distortional waves and light waves is still closer, in that distortional waves also exert a pressure equal to the energy per unit volume. But as I have shown in a paper on "Pressure Perpendicular to the Shear Planes in Finite Pure Shears, etc." (*ante*, p. 546), the attempt was faulty, and a more correct treatment of the subject only shows that there is probably a pressure. We cannot say more as to its magnitude than that if it exists it is of the order of the energy per unit volume.

When a beam is travelling through a material medium we may, perhaps, account for the angular momentum in it by the following considerations. On the electromagnetic theory the disturbance at any given point in a circularly polarised beam is a constant electric strain or displacement f uniformly revolving with angular velocity $\dot{\theta}$. In time dt it changes its direction by $d\theta$.

This may be effected by the addition of a tangential strain $f d\theta$; or the rotation is produced by the addition of tangential strain $f \dot{\theta}$ per second, or by a current $f \dot{\theta}$ along the circle described by the end of f . We may imagine that this is due to electrons drawn out from their position of equilibrium so as to give f , and then whirled round in a circle so as to give a circular convection current $f \dot{\theta}$. Such a circular current of electrons should possess angular momentum.

Let us digress for a moment to consider an ordinary conduction circuit as illustrating the possession of angular momentum on this theory. Let the circuit have radius a and cross-section s , and let there be N negative electrons per unit volume, each with charge e and mass m , and let these be moving round the circuit with velocity v . If i is the total current, $i = N s v e$. The angular momentum will be

$$N s 2\pi a \cdot m v a = 2\pi a^2 i m / e = 2A i m / e = 2M m / e,$$

where A is the area of the circuit and M is the magnetic moment. This is of the order of $2M/10^7$.

It is easily seen that this result will hold for any circuit, whatever its form if A is the projection of the circuit on a plane perpendicular to the axis round which the moment is taken and if $M = A i$. If we suppose that a current of negative electrons flows round the circuit in this way and that the reaction while their momentum is being established is

on the material of the conductor, then at make of current there should be an impulse on the conductor of moment $2M/10^7$. If the circuit could be suspended so that it lay in a horizontal plane and was able to turn about a vertical axis in a space free from any magnetic field, we might be able to detect such impulse if it exists. But it is practically impossible to get a space free from magnetic intensity. If the field is H , the couple in the circuit due to it is proportional to HM . It would require exceedingly careful construction and adjustment of the circuit to ensure that the component of the couple due to the field about the vertical axis was so small that its effect should not mask the effect of the impulsive couple. The electrostatic forces, too, might have to be considered as serious disturbers.

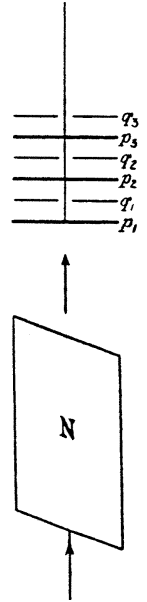
Returning to a beam of circularly polarised light, supposed to contain electrons revolving in circular orbits in fixed periodic times, the relations between energy and angular momentum are exactly the same as those in a revolving shaft or tube, and the angular momentum transmitted per second per square centimetre is $E\lambda/2\pi = P\lambda/2\pi$, where P is the pressure of the light per square centimetre on an absorbing surface.

The value of this in any practical case is very small. In light pressure experiments, P is detected by the couple on a small disc, of area A say, at an arm b and suspended by a fibre. What we observe is the moment APb . If the same disc is suspended by a vertical fibre attached at its centre and the same beam circularly polarised in both cases is incident normally upon it, according to the value suggested the torque is $AP\lambda/2\pi$.

The ratio of the two is $\lambda/2\pi b$. Now b is usually of the order of 1 cm. Put $\lambda = 6 \times 10^{-5}$, or, say, $2\pi/10^{-5}$, and the ratio becomes 10^{-5} .

It is by no means easy to measure the torque APb accurately, and it appears almost hopeless to detect one of a hundred-thousandth of the amount. The effect of the smaller torque might be multiplied to some extent, as shown in accompanying diagram.

Let a series of quarter wave plates, $p_1, p_2, p_3, \dots$, be suspended by a fibre above a Nicol prism N , through which a beam of light is transmitted upwards, and intermediate between these let a series of quarter wave plates, $q_1, q_2, q_3, \dots$, be fixed, each with a central hole for the free passage of the fibre. The beam emerges from N plane polarised. If N is placed so that the beam after passing through p_1 is circularly polarised, it has gained angular momentum, and therefore tends to twist p_1 round. The next plate q_1 is to be arranged so that the beam emerges from it plane polarised and in the original plane. It then passes through p_2 , which is similar to p_1 , and again it is circularly polarised and so exercises another torque. The process is repeated with q_2 and p_3 , and so on till the beam is exhausted. By revolving N through a right angle round the beam, the effect is reversed. But, even with such multiplications, my present experience of light forces does not give me much hope that the effect could be detected, if it has the value suggested by the mechanical model.



Mechanical Detection and Measurement of the Angular Momentum of Light

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The electromagnetic theory of the torque exerted by a beam of polarized light on a doubly refracting plate which alters its state of polarization is summarized. The same quantitative result is obtained by assigning an angular momentum of $\hbar$ ($-\hbar$) to each quantum of left (right) circularly polarized light in a vacuum, and assuming the conservation of angular momentum holds at the face of the plate. The apparatus used to detect and measure this effect

was designed to enhance the moment of force to be measured by an appropriate arrangement of quartz wave plates, and to reduce interferences. The results of about 120 determinations by two observers working independently show the magnitude and sign of the effect to be correct, and show that it varies as predicted by the theory with each of three experimental variables which could be independently adjusted.

ELECTROMAGNETIC FIELD THEORY

THE moment of force or torque exerted on a doubly refracting medium by a light wave passing through it arises from the fact that the dielectric constant $\mathbf{K}$ is a tensor. Consequently the electric intensity $\mathbf{E}$ is, in general, not parallel to the electric polarization $\mathbf{P}$ or to the electric displacement

$$\mathbf{D} = \mathbf{K}\mathbf{E} = \mathbf{E} + 4\pi\mathbf{P}$$

in the medium. The torque per unit volume produced by the action of the electric field on the polarization of the medium is

$$\mathbf{l} = \mathbf{P} \times \mathbf{E} = (\mathbf{D} \times \mathbf{E})/4\pi. \quad (1)$$

Following Sadowsky and Epstein¹ we may calculate this torque for a simple case as follows: Assume a doubly refracting medium of permeability unity and take as x , y , and z axes the principal axes of the tensor $\mathbf{K}$ for the light frequency in question. Denote the principal values of $\mathbf{K}$ by n_x^2 , n_y^2 and n_z^2 so that n_x , n_y and n_z will be the principal indices of refraction. Let the electric components of a plane light wave

propagated in the $+z$ direction be given by

$$\begin{aligned} \mathbf{E} &= A \cos \theta \cos (Z_1 + \Delta), \\ &\quad A \sin \theta \cos (Z_1 - \Delta), \quad 0, \\ \mathbf{D} &= \mathbf{K}\mathbf{E} = n_x^2 A \cos \theta \cos (Z_1 + \Delta), \\ &\quad n_y^2 A \sin \theta \cos (Z_1 - \Delta), \quad 0, \end{aligned} \quad (2)$$

where $Z_1 = \omega(t - nz/c)$; $n = (n_y + n_z)/2$,

$$\Delta = \pi z(n_y - n_x)/\lambda; \quad \omega = 2\pi c/\lambda.$$

For any value of z (in particular $z=0$) for which the wave plate thickness Δ/π is a whole number the light is plane polarized and E makes an angle $+\theta$ with the x axis.

The torque per unit volume at any point in the crystal is calculated from (2) according to (1) $\mathbf{l} = 0, \quad 0,$

$$(A^2/8\pi)(n_x^2 - n_y^2) \sin 2\theta \cos (Z_1 + \Delta) \cos (Z_1 - \Delta)$$

and the time average value of the z component is

$$(A/4\pi)^2(n_x^2 - n_y^2) \sin 2\theta \cos 2\Delta.$$

Integrating from z_1 to z_2 we get the torque per unit area on the crystal between these values:

$$L = -(A/4\pi)^2 n \lambda \sin 2\theta (\sin 2\Delta_2 - \sin 2\Delta_1). \quad (3)^2$$

This torque, which has hitherto been generally considered too small for experimental detection, was detected and measured in the present experiment.³

² J. H. Poynting, *Proc. Roy. Soc. A* **82**, 560 (1909), infers from an ingenious mechanical analogy that circularly polarized light should exert a torque equal to the light energy per unit volume times $\lambda/2\pi$ on unit area of a quarter-wave plate which makes the light plane polarized. This result, which neglects surface reflections, is contained in (3).

³ R. A. Beth, *Phys. Rev.* **48**, 471 (1935); also Annual Meeting of the Am. Phys. Soc., St. Louis, Missouri, January 1, 1936. A. H. S. Holbourn, *Nature* **137**, 31 (1936), also reports successful measurement of the effect.

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¹ A. Sadowsky, *Acta et Commentationes Imp. Universitatis Jurievensis* **7**, No. 1-3 (1899); **8**, No. 1-2 (1900). P. S. Epstein, *Ann. d. Physik* **44**, 593 (1914). The derivation is repeated here because the first articles are in Russian and relatively inaccessible, while in the last there seems to be a misprint in the result. I wish to thank Dr. Boris Podolsky for translating parts of Sadowsky's articles for me from the Russian. I am also deeply indebted to Professor A. Einstein not only for his advice in checking Professor Epstein's calculation but also for several interesting discussions about the experimental part of the work here reported.

QUANTUM THEORY

It is worth observing that (3) may be derived from the quantum theory by assigning an angular momentum or spin of $\hbar(-\hbar)^4$ to each photon of left (right) circularly polarized light, the spin axis being in the direction of propagation of the light.⁵

Consider an elliptically polarized light wave, propagated in the $+z$ direction in a vacuum, whose components in the directions of the principal axes of the ellipse (phase difference $=\pi/2$) have amplitudes X_0 and Y_0 . Let the components of the same wave, when resolved along an arbitrary pair of perpendicular x and y axes in the plane of the ellipse, be given by

$$\mathbf{E} = X \cos(Z + \Delta), \quad Y \cos(Z - \Delta), \quad 0, \quad (4)$$

where $Z = \omega(t - z/c)$ and 2Δ is the phase angle by which the y component lags behind the x component of the wave. It may be shown that⁶

$$XY \sin 2\Delta = X_0 Y_0, \quad X^2 + Y^2 = X_0^2 + Y_0^2. \quad (5)$$

If we now regard the wave (4) as the superposition of left and right circularly polarized components of amplitudes L and R , respectively, calculation shows that

$$L^2 - R^2 = XY \sin 2\Delta, \quad L^2 + R^2 = (X^2 + Y^2)/2.$$

The number of left circularly polarized photons transmitted per unit area per second is the Poynting energy flow $cL^2/4\pi$ divided by the energy per photon $2\pi\hbar\nu = 2\pi\hbar c/\lambda$, or $\lambda L^2/8\pi^2\hbar$. Multiplying by $\hbar$ gives $\lambda L^2/8\pi^2$, the angular momentum transmitted per unit area per second by the left circularly polarized component. Taking account of the similar expression for the right circularly polarized component, the angular momentum transmitted per unit area per second by the wave (4) is

$$\begin{aligned} M &= \lambda(L^2 - R^2)/8\pi^2 = \lambda XY \sin 2\Delta/8\pi^2 \\ &= \lambda X_0 Y_0/8\pi^2. \end{aligned} \quad (6)$$

It is well known that the linear momentum

transmitted per unit area per second is the energy flow divided by c , in this case:

$$(L^2 + R^2)/4\pi = (X^2 + Y^2)/8\pi = (X_0^2 + Y_0^2)/8\pi.$$

Thus the angular and linear momenta are proportional to the natural invariants (5) of the wave (4). For $X_0 = Y_0$ their ratio is $\lambda = \lambda/2\pi$, which is another form of Poynting's result.²

The value (6) of the angular momentum, which is independent of the quantum constant $\hbar$, can be derived from the linear momentum (Poynting vector divided by c) in a finite beam, but the elementary method given is more descriptive in bringing out the fact that the quantum theory leads to the same value (3) as the wave theory for the torque on a section of a crystal.

To show this equivalence we have now to obtain the angular momentum for a wave (2) *in the crystal* from the expression (6) for a wave *in a vacuum*, by placing the face of the crystal at $z = z_1$ (vacuum for $z < z_1$) and using the conservation of angular momentum and Fresnel's expressions for the amplitudes of the reflected and transmitted waves. In the reflected wave the y component of amplitude $Y(n_y - 1)/(n_y + 1)$ lags behind the x component of amplitude $X(n_x - 1)/(n_x + 1)$ by the phase angle 2Δ . We get the angular momentum transmitted through the face $z = z_1$ by calculating (6) for the direct and reflected waves in the vacuum (taking account of direction of transmission of each) and assuming the conservation of angular momentum at the face

$$M_1 = XY(n_x + n_y)\lambda \sin 2\Delta/4\pi^2(n_x + 1)(n_y + 1).$$

Using Fresnel's expressions for the amplitudes, we identify the transmitted wave *in the crystal* with (2) at $z = z_1$ by choosing

$$A \cos \theta = 2X/(n_x + 1); \quad A \sin \theta = 2Y/(n_y + 1);$$

$$\Delta = \Delta_1 = \pi z_1(n_y - n_x)/\lambda.$$

Hence the angular momentum transmitted by (2) at $z = z_1$ is

$$M_1 = (A/4\pi)^2 n \lambda \sin 2\theta \sin 2\Delta_1. \quad (7)$$

It can now be seen directly that the torque (3) is equal to the excess of the angular momentum (7) per unit area per second flowing into a section of the crystal at $z = z_1$ over that flowing out at $z = z_2$. Hence both the field and the

⁴ $\hbar = h/2\pi$.

⁵ A. E. Ruark and H. C. Urey, Proc. Nat. Acad. Sci. **13**, 763 (1927); Harnwell and Livingood, *Experimental Atomic Physics* (1933), p. 81; Brillouin, *Les Statistiques Quantiques*, Chap. III (1930).

⁶ See, e.g., Schuster and Nicholson, *Theory of Optics*, third edition, pp. 14, 15 (1924), especially Eqs. (15) and (17).

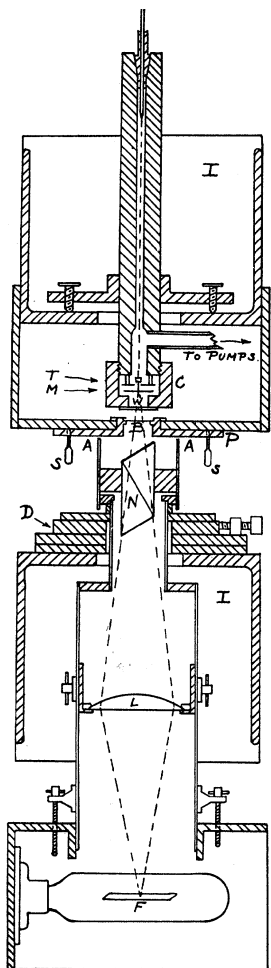


FIG. 1. Diagram of apparatus.

quantum theories predict the same value for the effect which has been measured in the present experiment.

APPARATUS

The basic idea used in detecting and measuring the effect⁷ is to observe the deflection of a quartz wave plate hung from a fine quartz fiber when suitably polarized light is sent through the plate. In order to increase the effect to be measured and to avoid as far as possible interfering effects due to mechanical and electrostatic disturbances,

⁷See Poynting, reference 2. A. Kastler, *Société des Sciences physiques et naturelles de Bordeaux*, Jan. 28, 1932. R. A. Beth, abstract, Boston Meeting, American Physical Society, *Phys. Rev.* 45, 296 (1934).

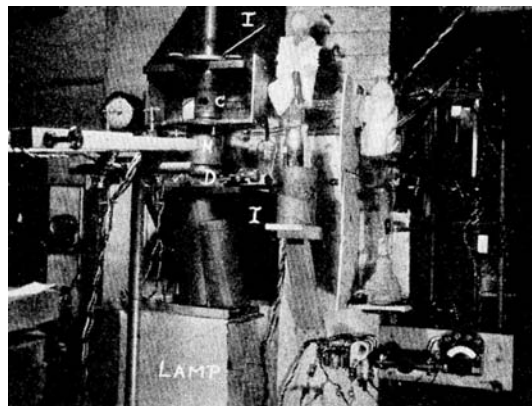


FIG. 2. Photograph of apparatus.

stray heating effects, radiometer and gas effects, and variations in light pressure, the apparatus was constructed as shown in Figs. 1, 2, 3, and 4. Some of the precautions taken to avoid these interferences may have been unnecessary, but they were sufficient.

The whole apparatus is supported on the heavy cast iron brackets I in Fig. 1, each 12"×12", which are bolted to a brick pier to avoid mechanical disturbances, especially torsional vibrations of a period comparable to that

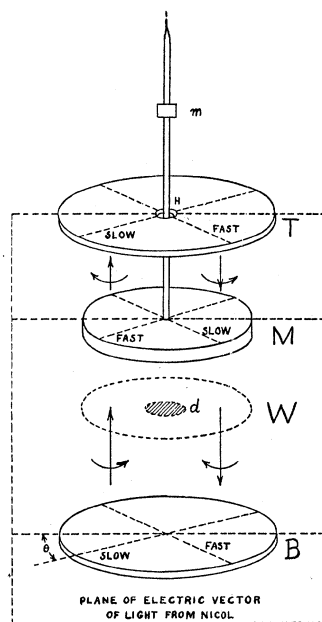


FIG. 3. Wave plate arrangement.

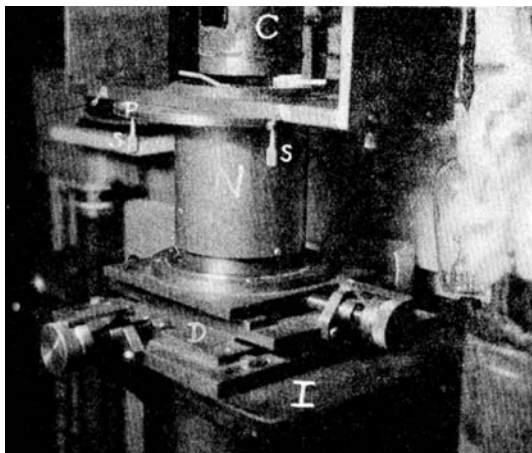


FIG. 4. Detail showing method by which light beam could be given lateral adjustment.

of the torsional pendulum formed by the one-inch circular wave plate M hanging at the bottom of the quartz fiber (about ten minutes). The fiber is about 25 centimeters long and hangs in a quarter inch hole bored lengthwise through a copper cylinder two inches in diameter. The upper end of the fiber is attached with a bit of flake shellac to a conical copper plug which was carefully ground in place and sealed with Apiezon grease. Thus the zero position as well as the amplitude of the torsional pendulum can be changed while the chamber is evacuated.

The hanging plate and the fixed plate above it, Fig. 3, are located in the cylindrical vacuum chamber C which is of copper, three inches in diameter with walls one-half inch thick. The heavy copper chamber is intended to eliminate unequal heating effects from the room as well as to shield off stray light and electrostatic disturbances.⁸

The chamber was evacuated through wide tubing by a high speed diffusion pump and three liquid-air traps. A two-stage diffusion pump and forepump provided the forevacuum for the high speed pump. According to the rated air calibration of the Western Electric ionization gauge, pressures below 10^{-6} mm of mercury existed in the chamber while the effect was being measured; no noticeable fluctuations appeared in the actual

measurements which might be ascribed to radiometer or gas effects.

Light from a three-millimeter tungsten ribbon filament F is focused by the fused quartz lens L , 10.7 centimeters in diameter, through a large Nicol prism N , the lower plate B (Fig. 3), the fused quartz window W , and the hanging plate M , on the reflecting layer of aluminum on the top of the upper plate T in the chamber. The maximum deviation of the light from the vertical was about 10° . W has a circular aperture one inch in diameter. A round one-quarter inch copper disk d fastened in the middle of W prevents light from passing through the hole H in the top plate T . With this arrangement no light energy reaches the fiber and most of the energy is reflected out of the vacuum chamber altogether, thus minimizing fluctuations in the position of the pendulum due to unequal heating of the fiber and light pressure on the small mirror m . These undesirable effects are found to be present if the shield d is removed.

The lower plate B is placed outside the vacuum chamber in a brass ring bushing held so it is free to turn in a brass frame attached to the upper cast iron bracket. The position of the lower wave plate may be read on a circular scale by means of a pointer attached to the bushing. A circular brass plate P attached to the under side of the bushing carries adjustable stops S , and the plate may be rotated from one stop to the other while observing the swing of the pendulum (with a telescope placed in front of the apparatus) by means of cords running in grooves on the edge of the plate P . Frame, plate P , cords, stops, pointer and circular scale are visible at the top of Fig. 4.

The ideas underlying the wave plate arrangement, Fig. 3, may be qualitatively understood as follows. The light coming upward through the plates is reflected by the aluminum player on the top side of T , passing downward again through the plates. For a certain wave-length λ_0 , B and T are quarter wave plates with their axes as shown at 90° to those of the hanging half-wave plate M . If light of wave-length λ_0 from the Nicol, plane polarized, at $\theta=45^\circ$, enters the bottom plate B , both direct and reflected beams will be circularly polarized in each space between the plates in the directions indicated by the

⁸ I wish to thank Professor G. P. Harnwell for this and many other helpful suggestions concerning the mechanical and optical design of the apparatus.

curved arrows. Thus the angular momentum delivered per second to the half-wave plate M is almost four times what could be obtained by simply absorbing the same amount of circularly polarized light.

For wave-lengths λ different from λ_0 it will be seen that, because of the alternate advancing and retarding of each component with respect to the other, light in both direct and reflected beams is plane polarized at the initial angle θ at the bottom of B , the middle of M , and the top of T . If the plates vary somewhat from the exact specifications these levels of plane polarization will be correspondingly shifted. In general the light will be elliptically polarized in each space between the plates. Careful consideration shows that the angular momentum delivered to M is in the same sense as that for λ_0 for all wave-lengths λ for which

$$(n_y - n_x)\lambda / \lambda < 2(n_y - n_x)\lambda_0 / \lambda_0$$

or, approximately, $\lambda > \lambda_0/2$.

If the lower plate B is rotated 90° from the position shown, the torque due to light of wave-length λ_0 will be just reversed. For this position the contributions to the torque of other wave-lengths is not easily considered qualitatively, but detailed calculation shows, as might be expected, that the integrated torque for all values of λ used is smaller in magnitude and opposite in direction compared to the torque similarly calculated for the position shown in Fig. 3. The torque is a continuous function of the position angle of the bottom plate, and its value can, with some labor, be calculated for each position.

The design wave-length λ_0 is determined* on the basis of (3) by noting that A^2 is practically proportional to the energy in the light at each wave-length, and that λ enters as a factor. The energy distribution has a rather sharp peak in the region just above 1.0μ for the tungsten filament temperatures to be considered. By a series of trial calculations $\lambda_0 = 1.2 \mu$ was chosen to make the light torque a maximum.

The actual plates used, of course, vary somewhat from the ideal specifications. The most accurate plates obtained were made to within a few percent of the specified thickness by the Bryden Company of Waltham, Massachusetts, whom I wish to thank for this excellent coopera-

tion. For the purposes of the exact calculations the plates were measured by means of a Babinet type compensator with sodium light. From the known values of the refractive indices of quartz⁹ the retardation was then calculated for each plate for all wave-lengths contributing to the effect.

Besides almost quadrupling the torque compared to what would be obtained if the light were simply absorbed, and furthermore reducing heating, radiometer, and gas effects in the vacuum chamber, this plate arrangement greatly reduced difficulties due to radiation pressure. The disk M cannot be mounted exactly at right angles to the fiber axis, nor can the light be projected exactly in a vertical direction. Thus the resultant light pressure integrated over the disk will not be quite vertical and will produce a "light pressure torque" unless its line of action lies in a vertical plane containing the fiber axis. Without special precautions this torque might very easily mask the effect to be detected and measured.¹⁰

A tungsten filament light source was chosen, at a sacrifice of light intensity compared to an arc, in order to keep the resultant light pressure on the disk M as steady as possible, both in position and magnitude, and to give a steady and reproducible spectral energy distribution for purposes of calculation. The undesirable resultant light pressure on the disk M is decreased in the plate arrangement described, first, because most of the energy is allowed to pass through the disk and, secondly, because the pressure due to surface reflections and absorption for the upward beam is largely cancelled by the similar pressure for the beam reflected downward.

Finally, the moment arm for the residual light pressure can be varied by shifting the entire light beam parallel to itself. This is done by mounting the entire optical system up to $A-A$ in Fig. 1 on the double slide D , which in turn is supported in the lower of the two cast iron brackets I . The detail view, Fig. 4, shows the two micrometer screws by which the lateral

⁹ *International Critical Tables*, VI, 341-342.

¹⁰ Besides the smallness of the effect itself, the light pressure torque is perhaps the greatest difficulty in this experiment. I wish to thank Professor A. Kastler of the University of Bordeaux for a personal communication emphasizing this fact.

motion in two coordinates is produced. These are calibrated in thousandths of an inch so that any particular setting is easily repeated at a later time. By using this arrangement and leaving the lower wave plate B in a fixed position, the change in the sum of light torque, and light pressure, radiometer, and gas effect torque produced by a ten-percent change in filament current was measured for various settings of one of the double slide screws, the other being kept fixed. The results when plotted give a smooth curve crossing the zero torque line when the filament image lies across the middle of the window W as nearly as the eye can judge. This indicates that the spurious torques are probably of the nature anticipated and, in any event, can be held steady and made very small by appropriate setting of the double slide. In making the measurements of the light torque effect itself, the filament current was held constant to better than one percent, and the filament image was placed across the middle of the window. Care was also taken to use the same lengthwise portion of the filament in all cases.

The entire optical system can be rotated in its support on the double slide. Changes in the angle θ so produced are read on the circular scale visible in Fig. 4 on top of the double slide and below the Nicol housing N .

As described below, changes in the light torque effect corresponding to definite rotations of the bottom plate were measured and found to be reproducible for the same settings to within a few percent. When the bottom plate was removed, or when it was replaced by a fused quartz plate, rotation in the same manner produced no measurable effect. Hence it may be assumed that rotating the lower plate leaves the light energy, light pressure, heating, radiometer and gas effects sensibly unchanged. In other words, the apparatus used successfully eliminates the interfering effects, and enhances the light torque effect to be measured.

MEASUREMENTS

The apparatus constructed according to the plan described showed the effect in the predicted direction and of about the predicted magnitude at the first trial on July 10, 1935. With this, the

first aim, that of *detecting* the effect, was accomplished. The second aim, that of *measuring* the effect, was carried out by a resonance method (see reference 7). One of the first series of measurements taken is shown in Fig. 5, which illustrates both the definiteness of the effect observed and the kind of minor fluctuations encountered.

At the top of Fig. 5 are given the readings $a_\nu (\nu=0, 1, 2, \dots)$ in centimeters on the scale (338 centimeters from the fiber axis) corresponding to the end points of the swing of the torsional pendulum. Tenths of millimeters on the scale could be estimated by means of the cross hairs in the telescope. The bottom plate B was in the position shown in Fig. 3 ($\theta=45^\circ$, right stop R) while the readings a_0 to a_4 were being taken, but was turned 90° to the left (to $\theta=-45^\circ$, to left stop L) at the instant at which the end point a_5 was read. The change in light torque produced by this rotation of B was such as to *oppose* the swing of the pendulum from a_5 to a_6 . At a_6 the plate B was turned back to R , the torque change again opposing the ensuing swing from a_6 to a_7 . The plate was turned similarly in "antiresonance" at a_7 and a_8 as indicated, but was then left in the position R when a_9, a_{10} and a_{11} were

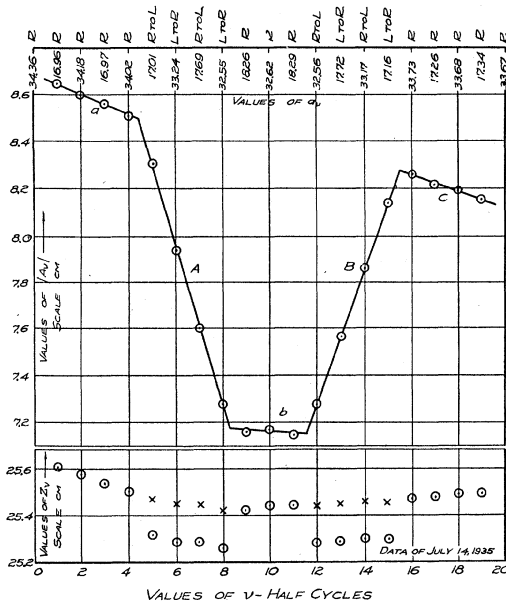


FIG. 5. Effect of light torque in resonance and out of resonance on the vibration amplitude of the torsion pendulum.

read. At a_{12} the plate was turned R to L , the change in torque now *aiding* the swing of the pendulum between a_{12} and a_{13} . At a_{13} the plate was turned L to R . This "resonance" turning was continued at a_{14} and a_{15} , and then the plate was again left in the position R for the drift readings a_{16} to a_{20} . For most of the readings taken another "antiresonance" series was added to the schedule described, and, for the readings taken by Mr. Harris, the "drift" periods were shortened while the "resonance" series in the middle of the schedule was lengthened to six half-cycles.

From the end-point readings taken according to such a schedule $4A_v = -a_{v-1} + 2a_v - a_{v+1}$ and $4Z_v = a_{v-1} + 2a_v + a_{v+1}$ were calculated. The absolute values of the "amplitudes" A_v are plotted in the upper part of Fig. 5 and the "zeros" Z_v (circles) in the lower part. For small values of damping (see "drift" slopes) the arithmetic means used differ by much less than the experimental error from the ideally correct geometric means for the quantities calculated.

From each such set of readings we wish to determine the value of the change in light torque on the hanging plate M when the bottom plate B is turned R to L or L to R . Since the torsional amplitude of the pendulum was at most a few degrees, we may assume that the light torque on M was a constant for a given position of the bottom plate B . We may consider a change in the position of the bottom plate to be equivalent to a change in the equilibrium position about which the torsional pendulum executes its oscillations. Let x be the corresponding change in scale reading. Careful consideration shows that, leaving damping out of account, the change in A_v per cycle must then be $2x$ during "resonance" or "antiresonance" sequences. With a small amount of damping, x is therefore very nearly equal to the average of the rates of change of A_v per half-cycle during "resonance" and "antiresonance." For example, in Fig. 5, the magnitudes of the slopes A and B , by least squares, are 0.341 and 0.289 centimeter per half-cycle, respectively. Hence $x = 0.315$ centimeter.

This method of taking readings and calculating affords several checks on the self-consistency of the sequence of readings a_v used to determine one value of x . For example, the damping slopes

a , b , and c should be approximately equal to $(A-B)/2$ in magnitude. Furthermore $Z_v + x/2$ during the "resonance" and "antiresonance" intervals (plotted as crosses) should form a smooth sequence with the values of Z_v during the drift intervals. Slight disturbances, such as between a_9 and a_{10} , may be taken into account in evaluating these diagrams.

The quartz fibers were made in a gas-oxygen blast flame. Of those which seemed straight and uniform over a length of a foot or so, the *finest* (small K)¹¹ was taken. K for the first pendulum, with which the measurements of Fig. 5 were taken, was 8.52×10^{-6} dyne-cm/radian, or 1.26×10^{-8} dyne-cm/scale centimeter. Hence the change in light torque in Fig. 5 was 3.97×10^{-9} dyne-cm. This first fiber was accidentally broken and most of the measurements were taken with the second fiber for which K was 10.1×10^{-6} dyne-cm/radian or 1.50×10^{-8} dyne-cm/scale centimeter. The period in the latter case was about nine minutes, so that most of the determinations of x required consecutive observation of end-points for practically two hours. I made about 80 determinations during July, August and September, 1935, and Mr. Wilbur Harris made about 40 determinations during October, November and December, 1935, after I had left Princeton. Each of us worked in the absence of the other, and each made determinations of the three types to be described. Mr. Harris used a different bottom plate from the one

¹¹ With regard to the choice of the torsion constant K (in dyne-centimeters per radian) for the fiber, the following is of interest. During a "resonance" interval the amplitude (neglecting damping) increases by an amount $2Nx$ in N cycles. For a given torque change, x is inversely proportional to K . The number of cycles in a given time is directly proportional to $\sqrt{K}$, assuming the moment of inertia of the pendulum is a constant. Hence the change in amplitude $2Nx$ in a given time is inversely proportional to $\sqrt{K}$. In other words, the time required to achieve a given increase in amplitude by the resonance method varies directly as $\sqrt{K}$, other factors being constant. Increasing K in this experiment would therefore involve an increase in the time required for each measurement, a disadvantage which was not present in the corresponding case of the Einstein-de Haas experiment, in which the impulse per half cycle was independent of the length of the cycle. Nevertheless, Professor Einstein has pointed out that it would also be worth while in this experiment to increase K considerably, in spite of the increase in time required to achieve a given change in amplitude, because then the ratio of the energy in the effect desired to the energy of random disturbances would become much more favorable. Furthermore, it is an advantage to have the end-point readings a_v come at shorter intervals insofar as the possible segregation of the larger random disturbances is concerned.

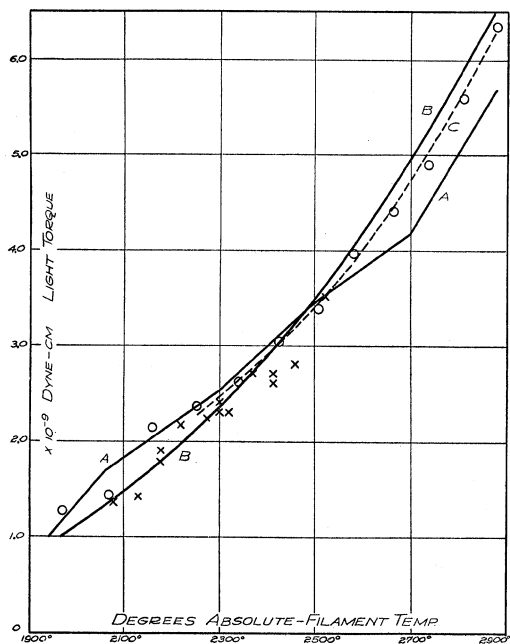


FIG. 6. Comparison of theory and experiment. Type I measurements: light torque as a function of filament temperature.

which I used, which seems to eliminate the possibility that the results, which were throughout consistent, could have been influenced by irregular patches of "infrared dirt" on the clear crystal quartz plates. Mr. Harris also investigated very thoroughly whether any effects were obtainable in the absence of the bottom plate, and found absolutely nothing. These checks were invaluable in substantiating the results of the present experiment, and Mr. Harris deserves much credit for the thoroughness and enthusiasm with which he carried out the tedious observations.

Three types of measurements, presented in Figs. 6, 8 and 9, were taken to show that the effect varies as required by the theory. The crosses represent my determinations and the circles those of Mr. Harris. Each point is derived from observations over a period of one and one-half to two hours as described in connection with Fig. 5. The smooth curves give the corresponding theoretical variation in each case.

For type I (Fig. 6) the torque change was measured for various filament temperatures as determined by the current through the lamp.

The bottom plate was turned between stops at $\theta = 45^\circ$ and $\theta = -45^\circ$, and the plane of polarization of the Nicol was at 45° to the axes of the plates as in Fig. 3. Mr. Harris' values (circles), taken consecutively, without shifting the position of the lamp, form the smoothest sequence obtained. The scattering of my values (crosses) taken with a different bottom plate is partly explained by the fact that I did not make these determinations consecutively, but shifted the lamp position a number of times for other readings. The total amount of light entering the plate system changes rather sharply with the position of the lamp in its socket and the setting of the double slide because of the presence of the small light shield d in the middle of the window W . The measurements previously reported¹² of this type, were made with a different lamp and fiber, and were made with a somewhat larger light shield on the window W . The values were therefore somewhat smaller than those shown in Fig. 6, but the magnitudes increased in a similar way with temperature.

The torque to be expected at various filament temperatures was calculated from (3) and the spectral energy distributions as follows. From the Poynting vector, the energy flow in the plate M is $A^2(n_x \cos^2 \theta + n_y \sin^2 \theta)c/8\pi$ or, practically, $A^2nc/8\pi = 2\pi c(A/4\pi)^2 n$ ergs/cm²-sec. Hence, if $J_\lambda d\lambda$ is the total energy flow (integrated over the area of the plate) in M in the wave-length range λ to $\lambda + d\lambda$, the net torque on the plate from the energy flow in this wave-length range will be $L_\lambda' d\lambda$ dyne-centimeters, where

$$L_\lambda' = \sin 2\theta_\lambda' (\sin 2\Delta_1 - \sin 2\Delta_2) \lambda J_\lambda' / 2\pi c.$$

Here $\tan \theta_\lambda'$ is, in general, the ratio of the amplitudes of the components of the light in the directions of the axes of the plates T and M . $\theta_\lambda' = \theta$ only when the position of the bottom plate B is such that the directions of its axes coincide with those for M and T . If the total energy flow entering the plate system in this wave-length range is $J_\lambda d\lambda$ ergs/sec., we set $J_\lambda' = J_\lambda F_{\lambda\eta}$ for each of the light beams contributing to the effect (enumerated by the index η), and compute $F_{\lambda\eta}$ using Edison Pettit's values for the reflectivity of aluminum (top surface of T)¹³ and the re-

¹² R. A. Beth, Phys. Rev. **48**, 471 (1935).

¹³ Edison Pettit, Pub. Astronom. Soc. Pacific **46**, 27 (1934).

fractive indices of quartz, which enter into Fresnel's expressions for the transmitted and reflected intensities at the plate surfaces. The absorption in the quartz plates may be neglected because the light has been filtered through the fused quartz lens L and the calcite and Canada balsam of the Nicol prism. From the theory for elliptically polarized light, the angles θ_λ' and the phases 2Δ may be calculated for each beam and each wave-length and for any position of the bottom plate, because the incoming light from the Nicol is plane-polarized, and the plate thicknesses, as measured by the compensator in wave plates for sodium light, were: $T=0.649$, $M=1.051$, $B_1=0.535$ (my torque measurements), $B_2=0.454$ (Harris' torque measurements). The whole torque on M is then

$$L = \int L_\lambda J_\lambda d\lambda \text{ dyne-centimeters,} \quad (8)$$

where the torque factors L_λ are given by

$$L_\lambda = \sum \lambda F_{\lambda\eta} \{ \sin 2\theta_\lambda' (\sin 2\Delta_1 - \sin 2\Delta_2) \}_{\lambda\eta/2\pi c} \quad \text{seconds.} \quad (9)$$

The torque factors L_λ were calculated for the sum η of the effects of the main beam going upward through M , the main beam reflected downward from the aluminized top surface of T , and for the reflections of these beams at the surfaces of B , W , M , and T , repeated surface

reflections being neglected. These values of (9) are shown in Fig. 7, the angles denoting the azimuthal position of the bottom plate; at 45° the direction of the "fast" axis of B coincides with the direction of the "slow" axis of M and the "fast" axis of T (see Fig. 3). The full curve refers to the calculations for B_2 , the dotted curve to those for B_1 .

Using a rocksalt spectrometer and thermocouple arrangement Mr. Harris compared the amount of radiation from a black body at a known temperature (radiation from the interior of a furnace made out of an alundum cylinder $2\frac{1}{2}$ inches in diameter and a foot long at 766°C and 784°C was used) with the amount of radiation coming through the Nicol in the same solid angle at intervals of 0.1μ for each of six different lamp currents. The energy factors J_λ were computed in absolute terms from these measurements using calculated values of black body radiation, Lambert's law, and constants of the optical system to determine the solid angle of the cone of light entering the plate system from the Nicol.

Replacing the integral in (8) by the corresponding sum for 0.1μ intervals from 1.0μ to 2.0μ inclusive, the change in total torque from $\theta=45^\circ$ to $\theta=-45^\circ$ for each of the six filament currents and for the bottom plate B_2 was obtained. These theoretical torque values are plotted against the temperatures, 1948° , 2069° , 2297° , 2506° , 2701° and 2879° absolute, corresponding to the filament currents used, in Fig. 6 and connected by straight lines A . The irregularity of this broken line A indicates the difficulties encountered in making the energy determinations. The principal sources of error are: The spectrometer was not set up especially for this purpose, but was originally designed for somewhat longer wave-lengths; the solid angle of the light cone is difficult to determine with accuracy; the neglected effect below 1.0μ is probably not exactly compensated by assuming that the Nicol actually polarizes all the light out to 2.0μ . There is reason to believe, from the energy measurements themselves, that the energies computed for the upper two temperatures are somewhat too low. The root-mean-square deviation of the measured torques (circles) from

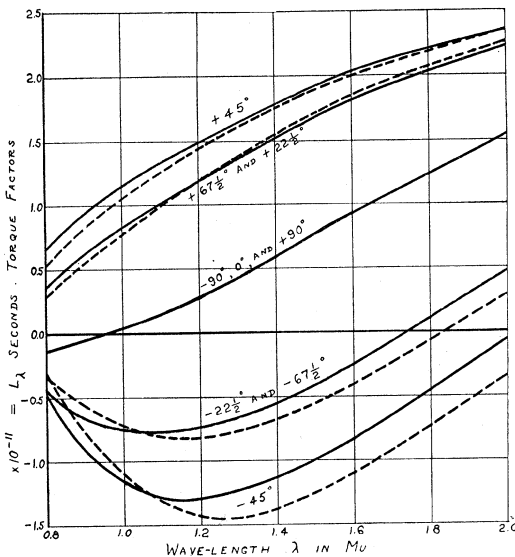


FIG. 7. Torque factor as a function of wave-length.

the computed values A is 0.29×10^{-9} dyne-centimeter.

A second attempt was made to get the values of the energy factors J_λ in absolute terms by extrapolating the tungsten emissivities given by Forsythe and Worthing¹⁴ (assuming the emissivity to be a linear function of temperature at each wave-length), multiplying by the black body radiation at the temperature of the filament, taking account of the solid angle of radiation collected from each point on the filament by the lens L , and making an allowance for the reflections from the surfaces of the bulb of the lamp, the lens, and the Nicol prism. The change in torque at each temperature was calculated from (8) as before, the integral in this case being replaced by the corresponding sum for 0.2μ intervals from 0.8μ to 2.0μ inclusive. The results must be reduced by about 15 percent to give the curve B in Fig. 6. This 15 percent may be caused by the absorption of the Canada balsam in the Nicol, by nonpolarization of the light near 2.0μ , and by inaccuracies in the emissivities used. The root-mean-square deviation of the circles from the curve B as drawn is 0.24×10^{-9} dyne-centimeter. It should be emphasized that the uncertainties in B leave the *absolute value* in considerably greater doubt than in the case of A .

An important conclusion from Fig. 6 is, however, that the *slope* and *shape* of the theoretically determined curves A and B agrees within the limits of error with the trend of the measured torque values. The trend of B relative to the torque values (measured for certain lamp currents) depends on the current-temperature calibration of the lamp which was extrapolated from one value at 2941° absolute by comparison with the calibration for a similar lamp, while the trend of A relative to the torque values measured does not depend on this calibration, the energy distributions having been measured for given lamp currents also. The dotted smooth curve C , empirically fitted to the upper nine of Harris' torque measurements, appears as a natural compromise between the trends of A and B . Half of the circles lie within one percent of this curve. Their root-mean-square deviation from the curve is 0.054×10^{-9} dyne-centimeter.

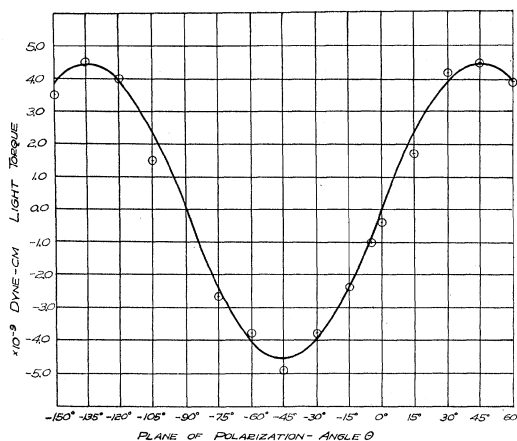


FIG. 8. Type II measurements: variation of light torque with angle between plane of polarization of light and the axes of the plates.

If we choose to think of C as an empirical calibration curve, this indicates that we may determine the temperature of the lamp in this range with a probable error of about 7° by measuring the torque with the accuracy attained by Mr. Harris, the lamp position being left unchanged.

The type I measurements may be regarded as a test of the factors $\lambda J_\lambda / 2\pi c$ in the calculation of the torque from (8) and (9).

For the type II measurements (Fig. 8) the lamp current, and therefore the light intensity, was maintained constant and the plate B_2 was rotated between the same stops as in type I, but the plane of polarization of the light was rotated to various angles θ with the axes of the plates. These measurements are intended to test the factor $\sin 2\theta$ in the formula for the torque. The root-mean-square deviation of the measured values from the computed sine curve shown is 0.34×10^{-9} dyne-centimeter. A small systematic deviation from the sine curve may be due to the difficulty encountered in lining up the plate axes accurately in setting up the apparatus. In setting the angle θ the whole lamp housing and optical system, including the Nicol, was rotated and the double slide reset as described. Because of the presence of the light shield d this causes a greater scattering of the measurements than would be obtained if the lamp could be left fixed in position.

The measurements shown were made by Mr. Harris. In my measurements of this type only a

¹⁴ Forsythe and Worthing, *Astrophys. J.* **61**, 151 (1925).

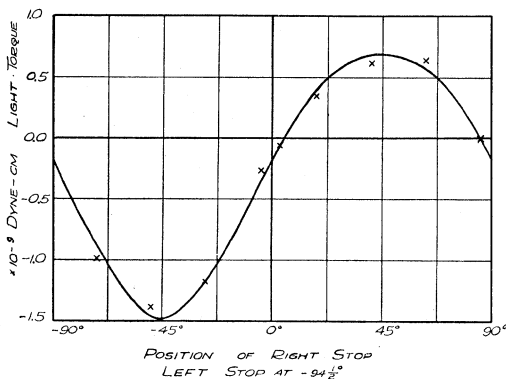


FIG. 9. Type III measurements: light torque as a function of position of the right stop.

quarter cycle of a sine curve was covered, but at 15° intervals. These points also form a smooth segment of a sine curve; though the root-mean-square deviation is considerably less, the test is not as critical and the result not as interesting as the full cycle curve shown.

For the type III measurements (Fig. 9) the light intensity was again maintained constant and the plane of polarization of the optical system was again put at 45° to the directions of the axes of plates *M* and *T* as in type I. The left stop was fixed at $-94\frac{1}{2}^\circ$ and measurements made for different positions of the right stop. Since the torque for *B*₁ at the left stop was constant, the curve shows how the light torque in this plate system varies with the position of the bottom plate. The elliptical polarization produced by the bottom plate depends on its position, and these measurements are, therefore, in the main a test of the phase factors ($\sin 2\Delta_1 - \sin 2\Delta_2$) in the formula. From the torque factors of Fig. 7 the smooth curve shown in Fig. 9 was calculated from the energy distribution. The root-mean-square deviation of the measured values of the torque from this curve is 0.073×10^{-9} dyne-centimeter. This very good agreement is again to be expected because the lamp position was left fixed. Mr. Harris also made measurements

of this type, but did not cover a whole cycle. They are in similar agreement with the predicted values.

CONCLUSION

The change in the angular momentum of polarized light in passing through a crystal plate has been detected and measured with the apparatus described. This conclusion rests on the fact that the measured effect agrees in sign and, within the limits of error, in magnitude with the effect as predicted both by the wave and by the quantum theories of light. In particular it has been shown that the effect varies as required by the theory with each of three experimental variables studied: filament temperature (light intensity), azimuth of the plane of polarization, and position of the bottom plate.

I am deeply indebted to Professor A. W. Duff of the Worcester Polytechnic Institute who first suggested this problem to me and who personally helped me with the purchase of apparatus in the preliminary stages of the investigation. I wish to thank the National Research Council for a Grant-in-Aid for the purchase of apparatus and the Palmer Physical Laboratory of Princeton for providing all the specially made apparatus and technical assistance as well as for the hospitality of the Laboratory. In particular I wish also to thank Professors Harnwell, Einstein, Condon, Robertson, Shenstone, Bleakney, and Dr. Barnes of Princeton for their willing helpfulness in discussing the work with me on numerous occasions. Dr. S. N. Van Voorhis and Messrs. C. W. Curtis and C. W. Lampson of Princeton helped me both with suggestions about experimental details and in actually taking readings. Mr. Curtis shared the work of taking readings with me for many weeks. The very valuable part of the work done by Mr. W. Harris has been noted in detail above. Finally, I want to thank Mr. A. H. S. Holbourn of the Clarendon Laboratory, Oxford, for his interesting correspondence with me about the problem.

Introduction to the atoms and angular momentum of light special issue

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Abstract

A very brief comment concerning the literature relating to the angular momentum of light is followed by a précis of a review article on the orbital angular momentum of light published in 1999. An outline is then given of the key developments since 1999 in the study of the angular momentum of light and of the related topic of optical vortices.

Keywords: Orbital angular momentum of light, optical vortices, atoms

1. Introduction

There is a vast physics literature related to the angular momentum of light. The great majority of it concerns spin angular momentum, discussed in the context of polarization, virtually all of it developed for idealized polarized plane waves. Sometimes a formal theoretical examination of the interaction of light and matter, particularly with a single atom, is written over a sum of plane waves. But, very often, the summation is something of an illusion and the final results are for a single plane wave of well defined wavevector k . A great proportion of the published work relates to the polarization selection rules of atomic transitions and, with the advent of laser trapping and cooling, to their exploitation for specific ends.

This does not mean that physicists believed that plane waves really existed. The approximation of a plane wave was used because it worked very well for most purposes. It has been known for a very long time, too, that a calculation of the angular momentum of a free electromagnetic field produced contributions which did not depend on photon spin. Attributing the term *orbital angular momentum* to those parts of the angular momentum of a light beam which did not arise from spin rarely proved necessary. Occasionally the insistence of angular momentum conservation, particularly in nuclear physics, dictated that a photon almost notionally carried off a certain number of units of $\hbar$, but in general the concept has not been much used.

A discussion of the behaviour of waves which were patently not plane, other than in the context of diffraction theory, did not really begin until the 1970s and early 1980s with the work of Nye and Berry [1, 2]. Their work on singularities and dislocations in wavefronts has led to a very considerable

literature. For screw dislocations, or phase singularities, this leads to an examination of the behaviour of *optical vortices*. Light beams possessing such a property can, of course, be generated in the laboratory using laser beams and they were first generated by Vaughan and Willetts in 1983 [3]. Even more recently, 1992, came the realization [4] that laboratory-generated laser beams could exist with a unique integer amount of orbital angular momentum, $\ell\hbar$ per photon. Only since then has serious thought been given in optical physics to the possibility of investigating such a property.

The literatures of optical vortices and of orbital angular momentum have stayed resolutely separate. At the lowest level this was because of the difference in the chosen language of the two areas of optical physics. It was also because the concept of angular momentum *per se* was not for some time strongly associated with optical vortices and, indeed, not all dislocations have an angular momentum: although angular momentum is conserved in a noninteracting light beam, it is not necessarily conserved in its vortex. Nevertheless, the discussion of stable beams consisting of a circularly symmetric field such as higher-order Laguerre–Gaussian modes lends itself rather readily to the use of orbital angular momentum, even though the beam does contain an optical vortex. When such a beam is polarized it lends itself even more so to the concept of *total angular momentum*. Generalizations about the separation of these two areas of interest are likely to be too severe, because the same effects are, as ever in physics, describable in more than one way. For example, in many cases the main property of a Laguerre–Gaussian beam exploited in an atom guiding or trapping experiment may well not be its degree of orbital angular momentum.

It was a particular pleasure for me, having been strongly identified with the work on orbital angular momentum, to learn that an Institute of Physics journal was to have a special issue on the subject of atoms and the angular momentum of light together with the role of vortices. A wide range of interests has been welcomed in the hope that the issue might play the role, in some senses at least, of an update of more formal reviews.

The editor of *Journal of Optics B: Quantum and Semiclassical Optics* suggested that I might care to write this introduction and that a possible form it might take would be to bring the *Progress in Optics* [5] review on orbital angular momentum, published in 1999 and co-authored by myself, Padgett and Babiker, up to date. However, I feel that the detailed underpinning required for a review article is inappropriate here. In consequence, what follows is an outline of published work since 1999 rather than a coherent exposition of it. It includes, where appropriate, papers on vortices, although my emphasis will inevitably be on angular momentum.

At the time of the *Progress in Optics* review, much had already been done. The essential paraxial approximation theory [4] of orbital angular momentum had been devised, as had a quantum mechanical equivalent [6]. The concept had been shown not to be an artefact of the paraxial approximation in a rigorous examination of an unapproximated solution to an arbitrary free field [7], although certain important problems still remained. A general way of producing Laguerre–Gaussian beams from a laser operating in high-order Gaussian modes by means of a cylindrical lens mode-converter [8] had already been devised, although this approach is often supplanted by their production with a hologram [9, 10].

Experimental evidence was produced of the existence of orbital angular momentum [11] and for the case of a Laguerre–Gaussian beam of $\ell = 1$ shown to be mechanically equivalent [12] to a circularly polarized beam where the spin angular momentum is by necessity unity, $\sigma = 1$. This experiment, of course, is the equivalent of Beth’s famous determination of $\hbar$ from the spin angular momentum of a circularly polarized beam. This work extended that of Ashkin *et al* [13] on optical tweezers to what may be called *optical spanners*.

The frequency shift known variously as the angular Doppler shift [14], the rotational frequency shift [15] or the shift arising from a rotating mode-converter [16] has been experimentally observed both for orbital and for spin and orbital angular momentum combined [17, 18]. The same shift was predicted as an azimuthal Doppler shift in the interaction of Laguerre–Gaussian laser light with atoms [19], in a body of theoretical work concerned with atoms interacting with such light beams [20–23].

Other work on atoms with such modes had also begun with the possibility of dark field excitation previously achieved by blocking the light [24], and the construction of novel optical traps [25]. Second-harmonic generation had been observed [26] with the frequency-doubled beam shown to possess twice the angular momentum of the pump beam, $2\ell\hbar$. The first theoretical description of the conservation of orbital angular momentum in soliton motion in nonlinear media had also been completed [27] essentially using the language of vortices, following the earlier work of Swartzlander and Law [28] on soliton propagation.

All of these aspects of the properties and behaviour of orbital angular momentum have been revisited and extended, in the time between the 1999 review and now. The topic, which had a slow and halting start, has undergone a rapid expansion. The areas of interest can be categorized as general theory, mode converters, optical tweezers, atoms, Bose–Einstein condensates and entanglement.

2. General theory

A series of analogues now exists which shows how similar orbital angular momentum is to spin angular momentum. Allen *et al* [29] devised the equivalent of Jones matrices, which describe the orbital angular momentum as it passes through an system of mode-order preserving optical elements in terms of the constituent Hermite–Gaussian modes and developed the matrices to account for light that has both spin and orbital angular momentum. Immediately [30], a similar matrix was described in terms of Laguerre–Gaussian modes and the approach was used to construct a variable-phase mode converter. This strong analogy between spin and orbital angular momentum enabled Padgett and Courtial [31] to propose a sphere for light beams in orbital angular momentum states which is precisely equivalent to the Poincaré sphere, positions on which represent the polarization state of the light. This equivalent sphere was used to interpret the rotational frequency shift of light beams with orbital angular momentum.

An $SU(2)$ basis for the proposed analogue of the Poincaré sphere was presented by Agarwal [32], who generalized the work to include quantized as well as partially coherent beams with orbital angular momentum. He also demonstrated that the $SU(N)$ version of the theory could be used for higher-order Laguerre–Gaussian beams. Ponomarenko [33] introduced a new class of partially coherent beams which carry optical vortices. He showed that any such beam can be represented as an incoherent superposition of fully coherent Laguerre–Gaussian modes of arbitrary order with the same azimuthal index, the same family examined by Agarwal. He investigated their free-space propagation properties. Serna and Movilla [34] extended the consideration of coherent beams with orbital angular momentum to partially coherent beams in terms of two elements of the beam matrix. They considered a family of more general fields and the relationship between the twist of the beam and the orbital angular momentum. Simon and Agarwal [35] present a phase-space description to derive a closed form expression for the Wigner function for Laguerre–Gaussian modes and to show that it is as compact as the one for Hermite–Gaussian modes.

The principle of operation of a new class of optical devices operating in quadratic nonlinear media, that mix wavefront dislocations in focused beams and produce certain patterns of bright spatial solitons, was described by Torner *et al* [36]. They argued that the orbital angular momentum of the light beam is central to this behaviour. Self-trapped necklace-ring beams that carry and conserve angular momentum were demonstrated by Soljacic and Segev [37], who showed that such beams can have a fractional ratio of angular momentum to energy.

The local linear momentum density in Laguerre–Gaussian beams has been used [38] to investigate classically the trajectory of the Poynting vector for light of any polarization,

while some difficulties of interpretation of the local angular momentum density are discussed. The local angular momentum density plays a vital role in the determination of what value of angular momentum determines the nature of an interaction with matter. Spin angular momentum is always intrinsic and no matter how, or where, an interaction takes place the angular momentum involved is $\hbar$ per photon. Berry [39] showed that, contrary to popular belief, orbital angular momentum is independent of the beam axis provided it is reasonable and possible to stipulate a direction for which the transverse momentum current is exactly zero. It is tempting, therefore, to say that orbital angular momentum is also always intrinsic, but O'Neill *et al* [40] have shown experimentally that it may be both extrinsic and intrinsic.

The well known spin-dependent linear and angular transverse shifts of the centre of gravity of partially and refracted light beams have been shown theoretically to have a non-spin-dependent analogue by Fedoseyev [41]. For a Laguerre–Gaussian beam the linear transverse shift is interpreted as a transformation of the orbital angular momentum at reflection and refraction: a parallel is drawn between this shift and the Goos–Hanchen shift.

Under the broad heading of general theory we may also include several other papers which appear in this issue. Desyatnikov and Kivshar (pages S58–S65) describe ring-profile optical solitons in nonlinear media whose propagation behaviour depends upon their orbital angular momentum. Piccarillo *et al* (pages S20–S24) investigate the competition between spin and orbital angular momentum in liquid crystals both theoretically and experimentally. What I believe to be the first totally QED approach to this topic has been applied by Dávila Romero *et al* (pages S66–S72) to represent the photonics of beams with orbital angular momentum interacting with a nonlinear medium. They apply the theory to second-harmonic generation as an illustration. Tiwari (pages S39–S46) maps out a philosophical and rather speculative view of relativity, entanglement and the reality of photons and proposes a laboratory experiment to test the consequences which would arise from a vortex structure for the photon.

Finally a very significant contribution to the general theory of this area is to be found in this issue. Barnett (pages S7–S16) introduces the use of angular-momentum flux as a natural description of angular momentum carried by light and solves a number of outstanding problems. Most important is the fact that the angular-momentum flux of a light beam about its direction of propagation can be separated into spin and orbital parts and that this separation is not only gauge invariant, but does not rely on the paraxial approximation. Remarkably, this gauge invariant separation and the fluxes of both spin and orbital angular momentum reduce to the known paraxial forms in the appropriate limit. The title of his paper suggests the most functional title in the future for this area of physics, namely *optical angular momentum*.

3. Mode converters

Perhaps not surprisingly, given the effectiveness of cylindrical lens and hologram mode converters, this area of the subject has not seen a very strong degree of activity. However, Clifford *et al* [42] generated high-order Laguerre–Gaussian modes

from a simple external-cavity diode laser and the detailed performance of a cylindrical lens mode converter has been analysed by examining the output beam for imperfections in the implementation of the converter [43]. Nevertheless this issue of *J. Opt. B: Quantum Semiclass. Opt.* has a paper by Chávez-Cerda *et al* (pages S52–S57), who use a holographic technique to produce high-order Mattieu beams, the orbital angular momentum content of which is discussed in terms of Bessel beams.

4. Optical tweezers

The possibility of revolving a small particle by means of a light beam and of developing light-driven micromachines is creating a distinct literature of its own. Friese *et al* [44] pointed out that the difficulty of earlier measurements with orbital angular momentum was that it was difficult to develop high power because of overheating and unwanted axial forces, but that this could be overcome by the use of transparent particles. They showed that an optical torque can be induced on microscopic birefringent particles of calcite held by optical tweezers. The particles either became aligned with the polarization, or spun with constant rotation frequency. Because the particles were transparent they could be held in three-dimensional optical traps at very high power at rotation rates of more than 350 rps. Higurashi *et al* [45, 46] also achieved optically induced mechanical alignment of trapped birefringent micro-objects by use of linearly polarized light. Trapping and rotation were achieved by the same Gaussian beam. The micro-objects were aligned about the laser beam axis and their angular position could be smoothly controlled by rotating the plane of the electric field of the light.

Luo *et al* [47] describe an optical micromotor where the spinning particle consists of a two-bead linkage. The large bead is trapped in optical tweezers and rotates about the laser beam axis. The small bead is coated with gold or palladium and generates a rotational torque through the change of momentum from the gradient radiation pressure of the same laser. A laser power of 29 mW generated a rotation of ~ 160 rpm. They suggest that, apart from developing optically operated mechanical devices, such a system could be used to twist macromolecules or to generate shear forces in a medium at the nano-level.

O'Neil and Padgett [48] reported the trapping of metallic particles in inverted optical tweezers. The particles were loosely confined in three dimensions to an annular region just below the beam waist where gravity is counterbalanced by the scattering forces. Trapping efficiency was improved by the use of a Laguerre–Gaussian beam and off-axis particle rotation was observed, which is induced by the orbital angular momentum of the beam. The polarization of the light did not influence the motion of the particles. The same workers also show [49] that axial trapping is improved in optical tweezers when a Laguerre–Gaussian beam is used rather than a Gaussian. However, contrary to earlier suggestions, the lateral efficiency is unchanged. Friese *et al* [50] also visited this area in an approach which invoked the force constants of a trap modelled as a harmonic oscillator.

Galajda and Ormos [51] show how light-induced polymerization of light curing resins may be applied to

generate rotating particles several microns in size. They produce mechanical devices, consisting of multiple moving parts, driven by these rotors. Friese *et al* [52] also show how the torque they previously described may be harnessed to drive an optically trapped microfabricated structure.

Finally, Nieminen *et al* [53] argue that systematic application of electromagnetic scattering theory can provide a general theory of laser trapping and yield results missing from existing theory. This is particularly important for wavelength scale particles. They present calculations of forces and torques on trapped particles obtained from this approach.

In this issue there is evidence of further activity in papers by Volke-Sepulveda *et al* (pages S82–S89) on the mechanical transfer of orbital angular momentum from a high-order Bessel beam to trapped particles, and on the rotation of microscopic propellers in laser tweezers by Galajda and Ormos (pages S78–S81).

5. Atoms

The intrinsic shape of the intensity distribution of Laguerre–Gaussian modes, as well as their orbital angular momentum, make them ideal for interactions with atoms. It is not surprising therefore that there is seemingly a fast-growing literature associated with them. Morsch and Meacher [54] proposed an all-optical method for channelling and cooling a cloud of pre-cooled atoms making use of the light shifts induced by the laser mode tuned close to the atomic resonance. They showed that optical pumping and inelastic collisions of the atoms with the optical walls of the trap gave rise to cooling. Because this cooling offsets the heating due to random scattering of photons, the trap did not suffer any lifetime limitation. They presented the results of a semi-classical Monte Carlo simulation for a three-dimensional model. Simultaneously, Kuppens *et al* [55] presented experimental and theoretical studies of polarization-gradient cooling of neon atoms at the dark centre of a TEM₀₁* mode which, of course, has no angular momentum. The transverse motion within the annular mode is cooled by two-dimensional optical molasses. The results were found to be in qualitative agreement with a one-dimensional quantum Monte Carlo simulation of cooling in the presence of an external light-shift potential. They found that the focusing properties of Laguerre–Gaussian modes were not altered in the presence of the cooling beams. The same group reported [56] on the application of the mode as a strong dipole potential to guide a slow-atom beam. Here polarization gradient cooling generates a bimodal momentum distribution.

Lembessis [57] considered an atom moving in a linearly polarized mode and included the Roentgen term in the electric dipole interaction. He displayed terms which involve the coupling of the photon angular momentum with the atomic particle angular momentum. An investigation was carried out by Liu and Milburn [58] into the classical two-dimensional nonlinear dynamics of cold atoms in far-off-resonant Laguerre–Gaussian beams. They showed that chaotic developments exist provided $\ell > 1$, when the beam is periodically modulated. The atoms are predicted to accumulate on several ring regions when the system enters a regime of global chaos.

A fascinating experiment on optical pumping in cold caesium atoms [59] presented experimental results on the

transfer of orbital angular momentum from one transition to another at a quite different frequency. This transfer was observed using a nondegenerate four-wave mixing process as an indirect tool. The fact that the orbital angular momentum was not only preserved but conserved, suggests that radiation processes of higher order than dipole should demonstrate mode conversion as both orbital and spin selection rules become invoked.

Truscott *et al* [60] produced an optically written waveguide in an atomic vapour using Laguerre–Gaussian modes and these experimental results were subsequently explained by use of a full density matrix approach by Kapoor and Agarwal [61]. Anderson *et al* [62] returned to the problem more recently and considered a five-level system in detail. A high degree of theoretical sophistication has been applied [63] to study the effect of phase fluctuations of a driving laser field on the dissipative and dipolar forces of a two-level atom by means of the phase diffusion model. The effect of the phase fluctuations on orbital angular momentum transfer to atoms was investigated for both Laguerre–Gaussian and Bessel beams.

6. Bose–Einstein condensates

The degree of activity involving atoms makes it no surprise that Bose–Einstein condensates (BECs) should also be addressed. Bolda and Walls [64] proposed a method of simultaneously taking a ground state to a different atomic hyperfine state and to a vortex trap state. This would be accomplished by use of one laser having a Laguerre–Gaussian mode profile. It is the vortex of the mode rather than the orbital angular momentum which would play the key role here just as in the work of Wright *et al* [65], who numerically investigated red-detuned Laguerre–Gaussian beams of various mode-index to produce suitable optical dipole toroidal traps for two-dimensional atomic BECs. The efficient optical generation of vortices in trapped Bose–Einstein condensates [66] was demonstrated numerically, while the creation of Bose–Einstein condensates was demonstrated in Rb in a specially designed hybrid, optical dipole and magnetic trap [67]. This trap allows the coherent transfer of matter waves into a pure optical potential waveguide, based again on the TEM₀₁* beam. It is also the spatial distribution of the mode which is dominant in the work of Tsurumi and Wadati [68]. They investigated ground state properties of toroidal condensates through the variational method. It was found that the condensate is attracted towards the axis of symmetry of the torus.

In this issue Alexander *et al* (pages S33–S38) analyse the structure and stability of vortices in hybrid atomic–molecular BECs. They predict new types of topological vortex state and show that their dynamics in the presence of losses are nontrivial. Also here is the work of Boussiakou *et al* (pages S25–S32), who develop a theory to determine the electric and magnetic properties of vortex states in BECs. The Lagrangian is symmetric in the electric and magnetic aspects of the problem and includes both the Aharanov–Casher and Roentgen interaction terms. Specific field distributions are evaluated for electric and magnetic dipole active BECs in a vortex state.

7. Entanglement

Perhaps the most exciting new aspect of orbital angular momentum and the most promising for fundamental work in the future is that of entanglement. The state of one particle in a two-particle entangled state defines the state of the second particle, while neither particle has a well defined state before measurement. An enormous literature exists for the entanglement of two-state quantum systems, in particular those involving the two orthogonal polarization states of photons. Mair *et al* [69] have demonstrated, by means of a parametric down-conversion experiment, that modes carrying orbital angular momentum can also show entanglement. As Laguerre–Gaussian modes can be used to define an infinite-dimensional discrete Hilbert space, this approach offers the possibility of entanglement involving many orthogonal quantum states. Consequently, it may be possible to resolve some aspects of long-standing arguments and discussion concerning the results on polarization states. Franke-Arnold *et al* [70] have investigated theoretically the orbital angular momentum correlation of a photon pair created in a spontaneous parametric down-conversion process. They show how the conservation of orbital angular momentum results from phase-matching in the nonlinear crystal. The same workers [71] calculate the anticipated correlation between measurements of the orbital angular momentum of the signal and idler beams of parametric down-conversion. The orbital angular momentum is measured by use of a hologram and displacement of the hologram with respect to the beam axis allows the measurement of superpositions of Laguerre–Gaussian modes. The correlations between such superposition modes show entanglement and could be used for Bell-type tests of nonlocality. In this issue Vaziri *et al* (pages S47–S51) examine superpositions of the orbital angular momentum for application in entanglement experiments in a not dissimilar way.

In an interesting paper by Arnaut and Barbosa [72] the orbital and spin angular momentum of single photons and entangled pairs of photons generated by parametric down-conversion was investigated. However, Eliel *et al* [73] show that the crucial role played by crystal symmetry in their work cannot be validated.

An even newer aspect of entanglement is given in this issue in the paper by Muthukrishnan and Stroud (pages S73–S77), who consider the exchange of spin and orbital angular momenta between a circularly polarized Laguerre–Gaussian beam and a single atom trapped in a two-dimensional harmonic potential. Spin and orbital angular momentum are individually conserved and result in the entanglement of the internal and external degrees of freedom of the atom.

8. Conclusions

The concepts of orbital angular momentum and of optical vortices have extended over a large number of aspects of optical physics. To what extent this can be expected to continue, and to what extent it will ultimately prove to be important, remains to be seen. Over a decade, the foundations have been put into place. My guess is that the subject will continue to grow, although it may become increasingly technical, and that

we are still a long way from appreciating its full impact. I should like to think that this special issue would make a positive contribution to that process. There would seem to be no reason why not. It contains a mixed bag of papers, some inevitably purely transitory, but nevertheless including several which I suspect will pass the test of time. If so, that is probably more than anyone could have reasonably hoped.

Amusingly enough for me I find, almost by accident, that following ‘the observation of the dynamical inversion of the topical charge of an optical vortex’ by Molina-Terriza *et al* [74], Padgett and I have cited their reference and the word ‘vortex’ in our short paper published here. If such inveterate users of the formalism of orbital angular momentum can begin to combine the separate literatures, however frail the link, perhaps there is some possibility that the separate literatures will begin to coalesce.

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